

Q. 1 – Q. 5 carry one mark each.

- Q.1 Choose the most appropriate phrase from the options given below to complete the following sentence.

India is a post-colonial country because

- (A) it was a former British colony
- (B) Indian Information Technology professionals have colonized the world
- (C) India does not follow any colonial practices
- (D) India has helped other countries gain freedom

- Q.2 Who _____ was coming to see us this evening?

- (A) you said
- (B) did you say
- (C) did you say that
- (D) had you said

- Q.3 Match the columns.

Column 1

- 1) eradicate
- 2) distort
- 3) saturate
- 4) utilize

Column 2

- P) misrepresent
- Q) soak completely
- R) use
- S) destroy utterly

- (A) 1:S, 2:P, 3:Q, 4:R
- (C) 1:Q, 2:R, 3:S, 4:P

- (B) 1:P, 2:Q, 3:R, 4:S
- (D) 1:S, 2:P, 3:R, 4:Q

- Q.4 What is the average of all multiples of 10 from 2 to 198?

- (A) 90
- (B) 100
- (C) 110
- (D) 120

- Q.5 The value of $\sqrt{12 + \sqrt{12 + \sqrt{12 + \dots}}}$ is

- (A) 3.464
- (B) 3.932
- (C) 4.000
- (D) 4.444

Q. 6 – Q. 10 carry two marks each.

- Q.6 The old city of Koenigsberg, which had a German majority population before World War 2, is now called Kaliningrad. After the events of the war, Kaliningrad is now a Russian territory and has a predominantly Russian population. It is bordered by the Baltic Sea on the north and the countries of Poland to the south and west and Lithuania to the east respectively. Which of the statements below can be inferred from this passage?

- (A) Kaliningrad was historically Russian in its ethnic make up
- (B) Kaliningrad is a part of Russia despite it not being contiguous with the rest of Russia
- (C) Koenigsberg was renamed Kaliningrad, as that was its original Russian name
- (D) Poland and Lithuania are on the route from Kaliningrad to the rest of Russia

- Q.7 The number of people diagnosed with dengue fever (contracted from the bite of a mosquito) in north India is twice the number diagnosed last year. Municipal authorities have concluded that measures to control the mosquito population have failed in this region.

Which one of the following statements, if true, does not contradict this conclusion?

- (A) A high proportion of the affected population has returned from neighbouring countries where dengue is prevalent
(B) More cases of dengue are now reported because of an increase in the Municipal Office's administrative efficiency
(C) Many more cases of dengue are being diagnosed this year since the introduction of a new and effective diagnostic test
(D) The number of people with malarial fever (also contracted from mosquito bites) has increased this year
- Q.8 If x is real and $|x^2 - 2x + 3| = 11$, then possible values of $|-x^3 + x^2 - x|$ include
(A) 2, 4 (B) 2, 14 (C) 4, 52 (D) 14, 52
- Q.9 The ratio of male to female students in a college for five years is plotted in the following line graph. If the number of female students doubled in 2009, by what percent did the number of male students increase in 2009?

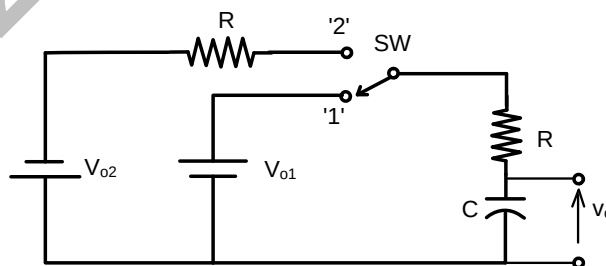


- Q.10 At what time between 6 a.m. and 7 a.m. will the minute hand and hour hand of a clock make an angle closest to 60° ?
(A) 6:22 a.m. (B) 6:27 a.m.
(C) 6:38 a.m. (D) 6:45 a.m.

END OF THE QUESTION PAPER

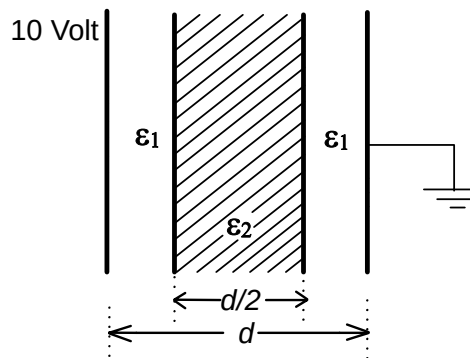
Q. 1 – Q. 25 carry one mark each.

- Q.1 Which one of the following statements is true for all real symmetric matrices?
 (A) All the eigenvalues are real.
 (B) All the eigenvalues are positive.
 (C) All the eigenvalues are distinct.
 (D) Sum of all the eigenvalues is zero.
- Q.2 Consider a dice with the property that the probability of a face with n dots showing up is proportional to n . The probability of the face with three dots showing up is _____.
- Q.3 Minimum of the real valued function $f(x) = (x - 1)^{2/3}$ occurs at x equal to
 (A) $-\infty$ (B) 0 (C) 1 (D) ∞
- Q.4 All the values of the multi-valued complex function 1^i , where $i = \sqrt{-1}$, are
 (A) purely imaginary. (B) real and non-negative.
 (C) on the unit circle. (D) equal in real and imaginary parts.
- Q.5 Consider the differential equation $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} - y = 0$. Which of the following is a solution to this differential equation for $x > 0$?
 (A) e^x (B) x^2 (C) $1/x$ (D) $\ln x$
- Q.6 Two identical coupled inductors are connected in series. The measured inductances for the two possible series connections are $380 \mu\text{H}$ and $240 \mu\text{H}$. Their mutual inductance in μH is _____
- Q.7 The switch SW shown in the circuit is kept at position '1' for a long duration. At $t = 0+$, the switch is moved to position '2'. Assuming $|V_{o2}| > |V_{o1}|$, the voltage $v_c(t)$ across the capacitor is



- (A) $v_c(t) = -V_{o2}(1 - e^{-t/2RC}) - V_{o1}$
 (B) $v_c(t) = V_{o2}(1 - e^{-t/2RC}) + V_{o1}$
 (C) $v_c(t) = -(V_{o2} + V_{o1})(1 - e^{-t/2RC}) - V_{o1}$
 (D) $v_c(t) = (V_{o2} - V_{o1})(1 - e^{-t/2RC}) + V_{o1}$

- Q.8 A parallel plate capacitor consisting two dielectric materials is shown in the figure. The middle dielectric slab is placed symmetrically with respect to the plates.



If the potential difference between one of the plates and the nearest surface of dielectric interface is 2 Volts, then the ratio $\epsilon_1 : \epsilon_2$ is

- (A) 1:4 (B) 2:3 (C) 3:2 (D) 4:1

- Q.9 Consider an LTI system with transfer function

$$H(s) = \frac{1}{s(s+4)}$$

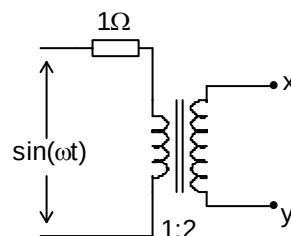
If the input to the system is $\cos(3t)$ and the steady state output is $A \sin(3t + \alpha)$, then the value of A is

- (A) 1/30 (B) 1/15 (C) 3/4 (D) 4/3

- Q.10 Consider an LTI system with impulse response $h(t) = e^{-5t} u(t)$. If the output of the system is $y(t) = e^{-3t} u(t) - e^{-5t} u(t)$ then the input, $x(t)$, is given by

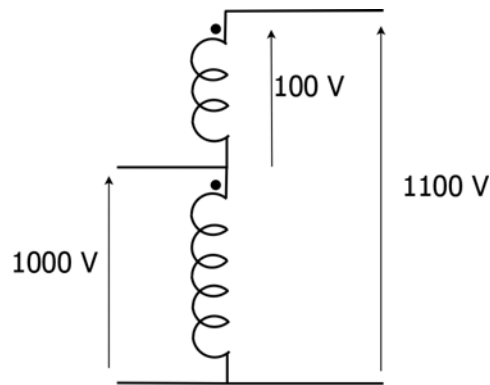
- (A) $e^{-3t} u(t)$ (B) $2 e^{-3t} u(t)$
(C) $e^{-5t} u(t)$ (D) $2 e^{-5t} u(t)$

- Q.11 Assuming an ideal transformer, the Thevenin's equivalent voltage and impedance as seen from the terminals x and y for the circuit in figure are



- (A) $2 \sin(\omega t)$, 4Ω (B) $1 \sin(\omega t)$, 1Ω
(C) $1 \sin(\omega t)$, 2Ω (D) $2 \sin(\omega t)$, 0.5Ω

- Q.12 A single phase, 50 kVA, 1000V/100 V two winding transformer is connected as an autotransformer as shown in the figure.



The kVA rating of the autotransformer is _____.

- Q.13 A three-phase, 4-pole, self excited induction generator is feeding power to a load at a frequency f_1 . If the load is partially removed, the frequency becomes f_2 . If the speed of the generator is maintained at 1500 rpm in both the cases, then
- (A) $f_1, f_2 > 50$ Hz and $f_1 > f_2$ (B) $f_1 < 50$ Hz and $f_2 > 50$ Hz
(C) $f_1, f_2 < 50$ Hz and $f_2 > f_1$ (D) $f_1 > 50$ Hz and $f_2 < 50$ Hz
- Q.14 A single phase induction motor draws 12 MW power at 0.6 lagging power. A capacitor is connected in parallel to the motor to improve the power factor of the combination of motor and capacitor to 0.8 lagging. Assuming that the real and reactive power drawn by the motor remains same as before, the reactive power delivered by the capacitor in MVAR is _____.
- Q.15 A three phase star-connected load is drawing power at a voltage of 0.9 pu and 0.8 power factor lagging. The three phase base power and base current are 100 MVA and 437.38 A respectively. The line-to-line load voltage in kV is _____.
- Q.16 Shunt reactors are sometimes used in high voltage transmission systems to
- (A) limit the short circuit current through the line.
(B) compensate for the series reactance of the line under heavily loaded condition.
(C) limit over-voltages at the load side under lightly loaded condition.
(D) compensate for the voltage drop in the line under heavily loaded condition.
- Q.17 The closed-loop transfer function of a system is $T(s) = \frac{4}{(s^2 + 0.4s + 4)}$. The steady state error due to unit step input is _____.

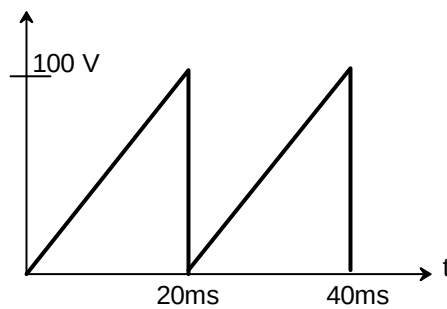
- Q.18 The state transition matrix for the system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u$$

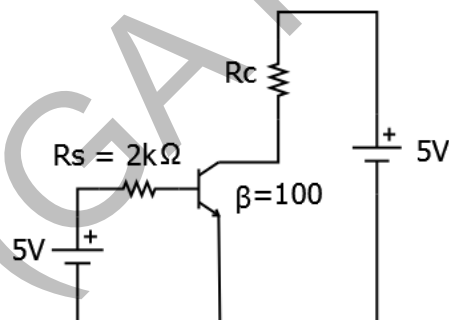
is

- (A) $\begin{bmatrix} e^t & 0 \\ e^t & e^t \end{bmatrix}$ (B) $\begin{bmatrix} e^t & 0 \\ t^2 e^t & e^t \end{bmatrix}$
(C) $\begin{bmatrix} e^t & 0 \\ t e^t & e^t \end{bmatrix}$ (D) $\begin{bmatrix} e^t & t e^t \\ 0 & e^t \end{bmatrix}$

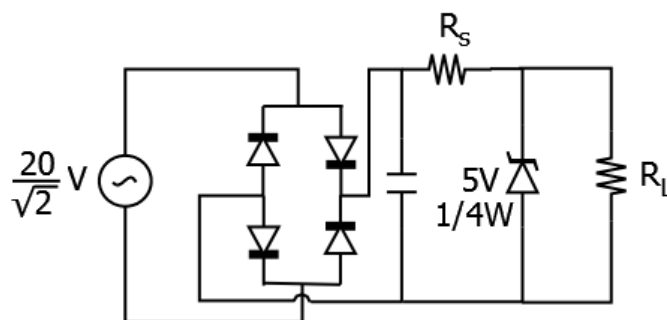
- Q.19 The saw-tooth voltage waveform shown in the figure is fed to a moving iron voltmeter. Its reading would be close to _____



- Q.20 While measuring power of a three-phase balanced load by the two-wattmeter method, the readings are 100 W and 250 W. The power factor of the load is _____.
- Q.21 Which of the following is an invalid state in an 8-4-2-1 Binary Coded Decimal counter
 (A) 1 0 0 0 (B) 1 0 0 1 (C) 0 0 1 1 (D) 1 1 0 0
- Q.22 The transistor in the given circuit should always be in active region. Take $V_{CE(sat)} = 0.2$ V, $V_{BE} = 0.7$ V. The maximum value of R_C in Ω which can be used, is _____.



- Q.23 The sinusoidal ac source in the figure has an rms value of $\frac{20}{\sqrt{2}}$ V. Considering all possible values of R_L , the minimum value of R_S in Ω to avoid burnout of the Zener diode is _____.



- Q.24 A step-up chopper is used to feed a load at 400 V dc from a 250 V dc source. The inductor current is continuous. If the 'off' time of the switch is 20 μ s, the switching frequency of the chopper in kHz is _____.

- Q.25 In a constant V/f control of induction motor, the ratio V/f is maintained constant from 0 to base frequency, where V is the voltage applied to the motor at fundamental frequency f . Which of the following statements relating to low frequency operation of the motor is TRUE?
- (A) At low frequency, the stator flux increases from its rated value.
 (B) At low frequency, the stator flux decreases from its rated value.
 (C) At low frequency, the motor saturates.
 (D) At low frequency, the stator flux remains unchanged at its rated value.

Q. 26 – Q. 55 carry two marks each.

- Q.26 To evaluate the double integral $\int_0^8 \left(\int_{y/2}^{(y/2)+1} \left(\frac{2x-y}{2} \right) dx \right) dy$, we make the substitution $u = \left(\frac{2x-y}{2} \right)$ and $v = \frac{y}{2}$. The integral will reduce to

- (A) $\int_0^4 \left(\int_0^2 2u \, du \right) dv$ (B) $\int_0^4 \left(\int_0^1 2u \, du \right) dv$
 (C) $\int_0^4 \left(\int_0^1 u \, du \right) dv$ (D) $\int_0^4 \left(\int_0^2 u \, du \right) dv$

- Q.27 Let X be a random variable with probability density function

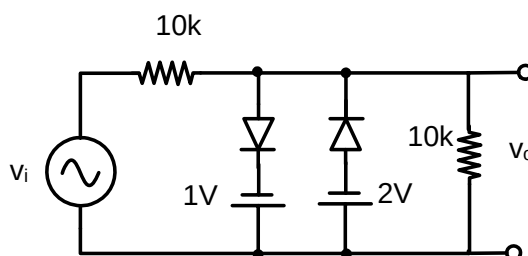
$$f(x) = \begin{cases} 0.2, & \text{for } |x| \leq 1 \\ 0.1, & \text{for } 1 < |x| \leq 4 \\ 0, & \text{otherwise.} \end{cases}$$

The probability $P(0.5 < X < 5)$ is _____.

- Q.28 The minimum value of the function $f(x) = x^3 - 3x^2 - 24x + 100$ in the interval $[-3, 3]$ is

- (A) 20 (B) 28
 (C) 16 (D) 32

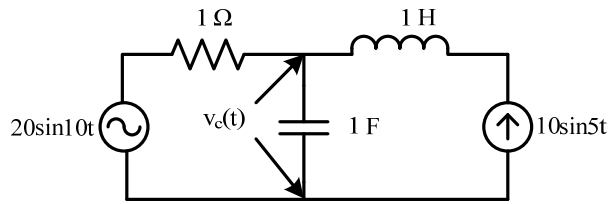
- Q.29 Assuming the diodes to be ideal in the figure, for the output to be clipped, the input voltage v_i must be outside the range



- (A) -1 V to -2 V (B) -2 V to -4 V
 (C) +1 V to -2 V (D) +2 V to -4 V

Q.30 The voltage across the capacitor, as shown in the figure, is expressed as

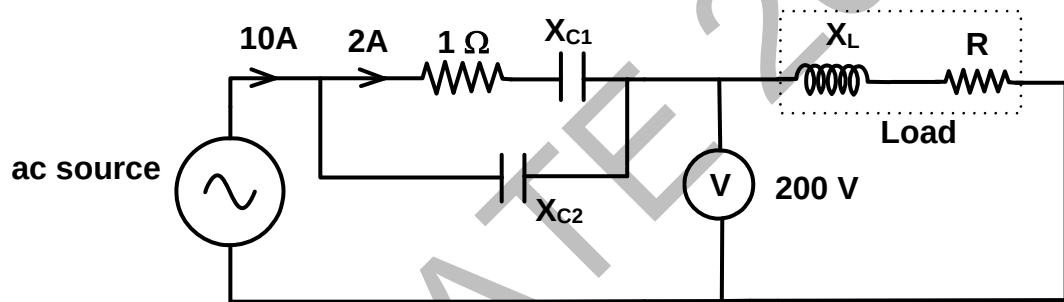
$$v_c(t) = A_1 \sin(\omega_1 t - \theta_1) + A_2 \sin(\omega_2 t - \theta_2)$$



The values of A_1 and A_2 respectively, are

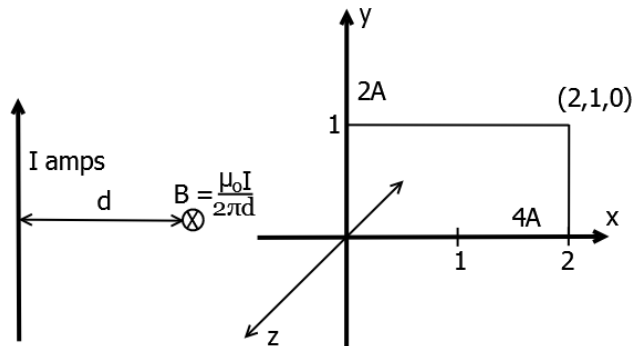
- (A) 2.0 and 1.98 (B) 2.0 and 4.20 (C) 2.5 and 3.50 (D) 5.0 and 6.40

Q.31 The total power dissipated in the circuit, shown in the figure, is 1 kW.



The voltmeter, across the load, reads 200 V. The value of X_L is _____.

Q.32 The magnitude of magnetic flux density ($\vec{B}$) at a point having normal distance d meters from an infinitely extended wire carrying current of I A is $\frac{\mu_0 I}{2\pi d}$ (in SI units). An infinitely extended wire is laid along the x-axis and is carrying current of 4 A in the +ve x direction. Another infinitely extended wire is laid along the y-axis and is carrying 2 A current in the +ve y direction. μ_0 is permeability of free space. Assume $\hat{i}$, $\hat{j}$, $\hat{k}$ to be unit vectors along x, y and z axes respectively.



Assuming right handed coordinate system, magnetic field intensity, $\vec{H}$ at coordinate (2,1,0) will be

- (A) $\frac{3}{2\pi} \hat{k}$ weber/m² (B) $\frac{4}{3\pi} \hat{i}$ A/m (C) $\frac{3}{2\pi} \hat{k}$ A/m (D) 0 A/m

Q.33 A discrete system is represented by the difference equation

$$\begin{bmatrix} X_1(k+1) \\ X_2(k+1) \end{bmatrix} = \begin{bmatrix} a & a-1 \\ a+1 & a \end{bmatrix} \begin{bmatrix} X_1(k) \\ X_2(k) \end{bmatrix}$$

It has initial conditions $X_1(0) = 1$; $X_2(0) = 0$. The pole locations of the system for $a = 1$, are

- (A) $1 \pm j0$ (B) $-1 \pm j0$ (C) $\pm 1 + j0$ (D) $0 \pm j1$

Q.34 An input signal $x(t) = 2 + 5\sin(100\pi t)$ is sampled with a sampling frequency of 400 Hz and applied to the system whose transfer function is represented by

$$\frac{Y(z)}{X(z)} = \frac{1}{N} \left(\frac{1 - z^{-N}}{1 - z^{-1}} \right)$$

where, N represents the number of samples per cycle. The output $y(n)$ of the system under steady state is

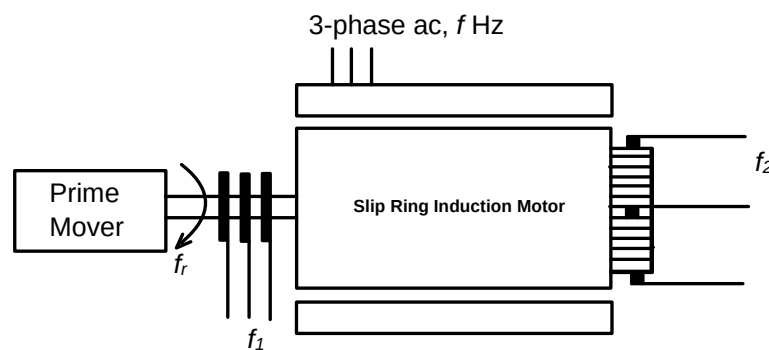
- (A) 0 (B) 1 (C) 2 (D) 5

Q.35 A 10 kHz even-symmetric square wave is passed through a bandpass filter with centre frequency at 30 kHz and 3 dB passband of 6 kHz. The filter output is

- (A) a highly attenuated square wave at 10 kHz.
(B) nearly zero.
(C) a nearly perfect cosine wave at 30 kHz.
(D) a nearly perfect sine wave at 30 kHz.

Q.36 A 250 V dc shunt machine has armature circuit resistance of 0.6Ω and field circuit resistance of 125Ω . The machine is connected to 250 V supply mains. The motor is operated as a generator and then as a motor separately. The line current of the machine in both the cases is 50 A. The ratio of the speed as a generator to the speed as a motor is _____.

Q.37 A three-phase slip-ring induction motor, provided with a commutator winding, is shown in the figure. The motor rotates in clockwise direction when the rotor windings are closed.



If the rotor winding is open circuited and the system is made to run at rotational speed f_r with the help of prime-mover in anti-clockwise direction, then the frequency of voltage across slip rings is f_1 and frequency of voltage across commutator brushes is f_2 . The values of f_1 and f_2 respectively are

- (A) $f + f_r$ and f (B) $f - f_r$ and f
(C) $f - f_r$ and $f + f_r$ (D) $f + f_r$ and $f - f_r$

- Q.38 A 20-pole alternator is having 180 identical stator slots with 6 conductors in each slot. All the coils of a phase are in series. If the coils are connected to realize single-phase winding, the generated voltage is V_1 . If the coils are reconnected to realize three-phase star-connected winding, the generated phase voltage is V_2 . Assuming full pitch, single-layer winding, the ratio V_1/V_2 is
- (A) $\frac{1}{\sqrt{3}}$ (B) $\frac{1}{2}$ (C) $\sqrt{3}$ (D) 2
- Q.39 For a single phase, two winding transformer, the supply frequency and voltage are both increased by 10%. The percentage changes in the hysteresis loss and eddy current loss, respectively, are
- (A) 10 and 21 (B) -10 and 21 (C) 21 and 10 (D) -21 and 10
- Q.40 A synchronous generator is connected to an infinite bus with excitation voltage $E_f = 1.3$ pu. The generator has a synchronous reactance of 1.1 pu and is delivering real power (P) of 0.6 pu to the bus. Assume the infinite bus voltage to be 1.0 pu. Neglect stator resistance. The reactive power (Q) in pu supplied by the generator to the bus under this condition is _____.
- Q.41 There are two generators in a power system. No-load frequencies of the generators are 51.5 Hz and 51 Hz, respectively, and both are having droop constant of 1 Hz/MW. Total load in the system is 2.5 MW. Assuming that the generators are operating under their respective droop characteristics, the frequency of the power system in Hz in the steady state is _____.
- Q.42 The horizontally placed conductors of a single phase line operating at 50 Hz are having outside diameter of 1.6 cm, and the spacing between centers of the conductors is 6 m. The permittivity of free space is 8.854×10^{-12} F/m. The capacitance to ground per kilometer of each line is
- (A) 4.2×10^{-9} F
(B) 8.4×10^{-9} F
(C) 4.2×10^{-12} F
(D) 8.4×10^{-12} F
- Q.43 A three phase, 100 MVA, 25 kV generator has solidly grounded neutral. The positive, negative, and the zero sequence reactances of the generator are 0.2 pu, 0.2 pu, and 0.05 pu, respectively, at the machine base quantities. If a bolted single phase to ground fault occurs at the terminal of the unloaded generator, the fault current in amperes immediately after the fault is _____.
- Q.44 A system with the open loop transfer function

$$G(s) = \frac{K}{s(s+2)(s^2+2s+2)}$$

is connected in a negative feedback configuration with a feedback gain of unity. For the closed loop system to be marginally stable, the value of K is _____

Q.45 For the transfer function

$$G(s) = \frac{5(s + 4)}{s(s + 0.25)(s^2 + 4s + 25)}$$

The values of the constant gain term and the highest corner frequency of the Bode plot respectively are

- (A) 3.2, 5.0 (B) 16.0, 4.0 (C) 3.2, 4.0 (D) 16.0, 5.0

Q.46 The second order dynamic system

$$\begin{aligned} \frac{dX}{dt} &= PX + Qu \\ y &= RX \end{aligned}$$

has the matrices P, Q and R as follows:

$$P = \begin{bmatrix} -1 & 1 \\ 0 & -3 \end{bmatrix} \quad Q = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad R = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

The system has the following controllability and observability properties:

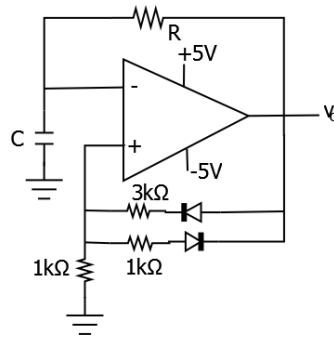
- (A) Controllable and observable
(B) Not controllable but observable
(C) Controllable but not observable
(D) Not controllable and not observable

Q.47 Suppose that resistors R_1 and R_2 are connected in parallel to give an equivalent resistor R . If resistors R_1 and R_2 have tolerance of 1% each, the equivalent resistor R for resistors $R_1 = 300 \, \Omega$ and $R_2 = 200 \, \Omega$ will have tolerance of

- (A) 0.5% (B) 1% (C) 1.2% (D) 2%

Q.48 Two ammeters X and Y have resistances of $1.2 \, \Omega$ and $1.5 \, \Omega$ respectively and they give full-scale deflection with 150 mA and 250 mA respectively. The ranges have been extended by connecting shunts so as to give full scale deflection with 15 A. The ammeters along with shunts are connected in parallel and then placed in a circuit in which the total current flowing is 15A. The current in amperes indicated in ammeter X is _____.

Q.49 An oscillator circuit using ideal op-amp and diodes is shown in the figure.



The time duration for +ve part of the cycle is Δt_1 and for -ve part is Δt_2 . The value of $e^{\frac{\Delta t_1 - \Delta t_2}{RC}}$ will be _____.

Q.50 The SOP (sum of products) form of a Boolean function is $\Sigma(0,1,3,7,11)$, where inputs are A, B, C, D (A is MSB, and D is LSB). The equivalent minimized expression of the function is

- (A) $(\bar{B} + C)(\bar{A} + C)(\bar{A} + \bar{B})(\bar{C} + D)$
- (B) $(\bar{B} + C)(\bar{A} + C)(\bar{A} + \bar{C})(\bar{C} + D)$
- (C) $(\bar{B} + C)(\bar{A} + C)(\bar{A} + \bar{C})(\bar{C} + \bar{D})$
- (D) $(\bar{B} + C)(A + \bar{B})(\bar{A} + \bar{B})(\bar{C} + D)$

Q.51 A JK flip flop can be implemented by T flip-flops. Identify the correct implementation.

- (A)
- (B)
- (C)
- (D)

Q.52 In an 8085 microprocessor, the following program is executed

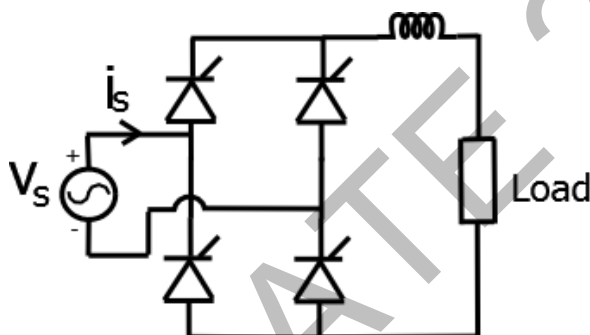
Address location – Instruction

2000H	XRA A
2001H	MVI B,04H
2003H	MVI A, 03H
2005H	RAR
2006H	DCR B
2007H	JNZ 2005
200AH	HLT

At the end of program, register A contains

- (A) 60H (B) 30H (C) 06H (D) 03H

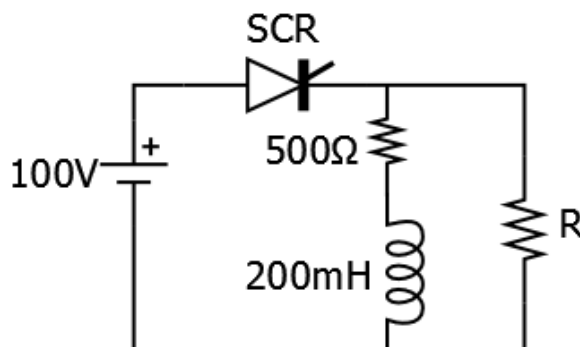
Q.53 A fully controlled converter bridge feeds a highly inductive load with ripple free load current. The input supply (v_s) to the bridge is a sinusoidal source. Triggering angle of the bridge converter is $\alpha = 30^\circ$. The input power factor of the bridge is _____.



Q.54 A single-phase SCR based ac regulator is feeding power to a load consisting of $5\ \Omega$ resistance and 16 mH inductance. The input supply is 230 V, 50 Hz ac. The maximum firing angle at which the voltage across the device becomes zero all throughout and the rms value of current through SCR, under this operating condition, are

- (A) 30° and 46 A (B) 30° and 23 A
(C) 45° and 23 A (D) 45° and 32 A

Q.55 The SCR in the circuit shown has a latching current of 40 mA. A gate pulse of $50\ \mu\text{s}$ is applied to the SCR. The maximum value of R in Ω to ensure successful firing of the SCR is _____.



END OF THE QUESTION PAPER

GATE 2014**Answer Keys for EE - Electrical Engineering**

Section	Q. No.	SESSION - H	
		Key / Range	Marks
GA	1	A	1
GA	2	B	1
GA	3	A	1
GA	4	B	1
GA	5	C	1
GA	6	B	2
GA	7	D	2
GA	8	D	2
GA	9	140 to 140	2
GA	10	A	2
EE	1	A	1
EE	2	0.13 to 0.15	1
EE	3	C	1
EE	4	B	1
EE	5	C	1
EE	6	35 to 35	1
EE	7	D	1
EE	8	C	1
EE	9	B	1
EE	10	B	1
EE	11	A	1
EE	12	545 to 555	1
EE	13	C	1
EE	14	6.97 to 7.03	1
EE	15	117 to 120	1
EE	16	C	1
EE	17	0 to 0	1
EE	18	C	1
EE	19	56 to 59	1
EE	20	0.78 to 0.82	1
EE	21	D	1
EE	22	22 to 23	1
EE	23	299 to 301	1

Section	Q. No.	SESSION - H	
		Key / Range	Marks
EE	24	31.0 to 31.5	1
EE	25	B	1
EE	26	B	2
EE	27	0.35 to 0.45	2
EE	28	B	2
EE	29	B	2
EE	30	A	2
EE	31	17.3 to 17.4	2
EE	32	C	2
EE	33	A	2
EE	34	C	2
EE	35	C	2
EE	36	1.22 to 1.32	2
EE	37	A	2
EE	38	D	2
EE	39	A	2
EE	40	0.10 to 0.12	2
EE	41	49.9 to 50.1	2
EE	42	B	2
EE	43	15300 to 15500	2
EE	44	5 to 5	2
EE	45	A	2
EE	46	C	2
EE	47	B	2
EE	48	9.9 to 10.3	2
EE	49	1.2 to 1.3	2
EE	50	A	2
EE	51	B	2
EE	52	A	2
EE	53	0.74 to 0.82	2
EE	54	C	2
EE	55	6055 to 6065	2